PRICE: A Certified Exploration Framework for Pricing Models That Shape Their Own Data

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***Abstract*—Corporate-bond dealers use pricing models to attract customers while managing the risk of trading with better-informed counterparties. These models learn from completed trades, but the prices they offer determine which customers trade and which data become available for future training. As a result, repeated retraining can converge to a stable pricing policy that falls short of the best achievable outcome. Improving this policy requires experimenting with alternative prices to learn how customers respond, but experimentation can sacrifice profit. Existing research studies model optimization and costly experimentation separately, leaving unresolved how to balance the cost of learning against the need for stable retraining. This paper develops PRICE, a framework for integrating the cost of exploration into pricing optimization while preserving retraining stability. To address this challenge, SafeD-PerfGD directs experiments toward price changes with the lowest impact on the dealer's objective and updates pricing policies only when stability can be mathematically guaranteed. Evaluation across two simulated economies and a public corporate-bond calibration confirms the theoretical results, passing all 36 preregistered checks and reproducing all ten calibration cells. Across 12 random seeds, no baseline achieves both lower median final error and lower median regret.**

I. Introduction

A. Problem Statement

Corporate-bond dealers must price bonds to attract customers while avoiding losses from trading with better-informed counterparties. Pricing models help balance these risks by learning from past transactions. However, the prices they recommend also determine which customers trade and what data become available for future training. This feedback can produce a stable pricing policy without reaching the performative optimum, which accounts for how pricing decisions change future data. In a public corporate-bond calibration, the gap between these outcomes amounts to 1.9–4.3% of the quoted spread [1].

Closing this gap requires experimenting with alternative prices to learn how customers respond. Yet experimentation can sacrifice immediate profit, and incorporating new estimates can destabilize retraining. The challenge is therefore to identify which experiments provide useful information at the lowest cost and when that information can safely improve the pricing model.

B. Current Approaches

Research on performative prediction studies models whose decisions influence their future training data. Perdomo et al. [2] establish conditions for convergence, while Miller et al. [3] show that a stable outcome need not be optimal. Subsequent work develops optimization algorithms, regret guarantees, and methods for estimating model-induced changes in data [4]–[11]. However, this literature does not fully account for the economic cost of the experimentation needed to estimate those changes.

A separate literature on online pricing, demand learning, dual control, and experimental design studies how to learn customer behavior while managing experimentation costs [12]–[26]. These methods, however, do not fully address how exploration interacts with the stability of repeated model retraining.

This gap is especially consequential in corporate-bond pricing. Costly experiments can reduce profitability, while updates based on new estimates can undermine retraining stability. A method that considers only one concern may either waste resources learning or converge reliably without discovering better prices. A unified approach must therefore balance the economic cost of exploration with the need to preserve stable retraining.

C. Proposed Approach

To address this challenge, this paper proposes PRICE, a three-stage framework that connects exploration cost, estimation accuracy, and retraining stability. Its first stage quantifies the cost of learning customer responses; the second identifies the cheapest directions in which to experiment; and the third uses these insights to improve pricing safely through SafeD-PerfGD.

First, PRICE establishes that, along any steady exploration policy, estimation variance multiplied by cumulative exploration cost equals $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$. This identity quantifies the cost of reducing uncertainty about market responses. Additional results establish lower bounds on exploration cost and show that knowing the market-response function does not eliminate this trade-off near the operating point.

Second, PRICE shows that experiments need not be distributed equally across all pricing directions. By concentrating exploration along directions where the dealer's objective is less sensitive to price changes, the framework reduces the cost of acquiring useful information.

Third, SafeD-PerfGD applies these results while preserving retraining stability. It explores along low-cost directions and updates prices only when a mathematical certificate guarantees

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contractive retraining, meaning successive updates continue to converge.

Evaluation across two simulated economies and a public corporate-bond calibration passes all 36 preregistered checks and reproduces all ten calibration cells. Across 12 random seeds, no baseline achieves both lower median final error and lower median regret.

D. Results Overview

Three results justify the framework as an improvement over current practice. The predicted cost of learning matches the measured cost to within about 1% (0.01062 and 0.01070 against 0.01056), so the price of experimenting can be computed in advance. Spreading experiments evenly across prices costs 1.53 times more than the optimal shape, as predicted (1.506). And the certified agent reaches the performative optimum with zero stability violations, while the fastest-learning baseline pays more than five times its regret to get there.

II. Background

A. The Dealer Market and the Retraining Loop

A dealer earns money on the spread h, the fee built into every trade. Some customers know more than the dealer about where the bond's price is heading, and trading with them loses money. A tighter spread attracts more trades of both kinds: informed flow arrives at a rate $\tau(h) = C_0 + C_1 e^{-ch}$ that rises as the spread shrinks, so the dealer's profit is a hill in h whose peak depends on how customers respond. Current practice is blind retraining: the dealer refits its quote to whatever trades it observed. Because the new quote changes the next day's trades, this is a feedback loop, and it settles only if the market responds less strongly than the dealer adjusts, a ratio we call the modulus m and require to be below one. Where it settles, the stable point h^*, we call the anchor. The performative optimum h_{PO}, the quote that would earn the most once the market's response is counted, lies at a tighter spread. The curvature γ_{PO}, how steeply profit falls as the quote moves away from the anchor, is the price per unit of experimenting.

B. Target Quantity – The Response Slope and Its Price

To leave the anchor safely, the dealer must learn a single number, the response slope ε: how much more informed flow one unit of tightening attracts. It can only be learned by moving the quote and watching the flow change. With d_t the distance of today's quote from the anchor and u_t the deliberate experiment,

$$d_{t+1} = -m d_t + u_t, \quad \tau_t = \tau(h^*) - \varepsilon d_t + \sigma\zeta_t \qquad (1)$$

The first part of (1) says that retraining pulls each deviation back toward the anchor while the dealer's experiments push it out; the second says that the observed flow reveals ε only through those deviations, blurred by noise of size σ. Every deviation that teaches the dealer something is also a trade at a worse price, a structure we term priced observation. The cost of learning can be counted in squared deviations, the deviation budget D_T, or in profit given up, the value budget C_T; the second is what a dealer actually pays. The estimate $\hat{\varepsilon}$ is the slope of a regression of observed flow on the quote.

C. Optimal Design and Certified Correction

This paper additionally builds upon two tools. Optimal experimental design [24], [25] asks which experiments yield the most information per unit of cost, and the van Trees inequality [26] bounds how accurate any estimate can be, even one that adapts as data arrive. Confidence sequences are intervals for ε that remain valid no matter how often they are checked; a certificate converts such an interval into a guarantee that retraining with the corrected quote still settles. It is required because acting on an estimate must never send the loop into oscillation.

III. Methods

A. Data Collection

All data comes from public sources. The corporate-bond leg uses a published calibration of a dealer market [1] covering ten cells, investment-grade and high-yield bonds crossed with five volatility regimes, built from 170 bonds with at least 12 months of returns. For each cell we recover the anchor h^*, the response slope ε^*, and the modulus m, and from them the curvature γ_{PO} and the gap between the anchor and the optimum. The simulated leg uses the dealer market of Section III-C and a second, deliberately simpler economy, a seller whose price shifts demand with gain g, to test whether the theory depends on the dealer setting.

B. The Exchange-Rate Identity and Minimax Floors

Near the anchor, profit lost is proportional to the squared distance from it, so the value budget is $C_T \approx \frac{1}{2}\gamma_{\mathrm{PO}} D_T$. Information about ε also grows with that same squared distance, so the remaining uncertainty is proportional to σ^2/D_T. Multiplying the two, the distance cancels, and for the regression estimate on any steady experiment schedule,

$$\mathrm{Var}(\hat{\varepsilon}) \times \frac{1}{2}\gamma_{\mathrm{PO}} D_T = \frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2(1 + O_P(1/T)) \qquad (2)$$

Uncertainty times cost is a constant of the market. Experimenting harder buys knowledge faster, but at the same exchange rate: halving the uncertainty always costs twice as much profit, whatever schedule is used. The floors say that no strategy, including ones that adapt as they go, escapes this rate. Under a budget on squared deviations the floor equals the exchange rate exactly; under a budget on profit the floor is $\frac{1}{27}\gamma_{\mathrm{PO}}\sigma^2$, within a factor 13.5 of what the best strategy achieves.

C. The Simulated Dealer Market

We simulate the dealer of Section II-A on a live deploy–observe–fit–correct pipeline at moduli $m \in \{0.3, 0.6, 0.85\}$, exploration amplitudes $s_e \in \{0.05, 0.15\}$, and horizons of $T = 4000$ deployments for the agent and $T = 4500$ for the ablation grid. The theory assumes a smooth, curved profit hill near the anchor, experiments that do not look into the future, Gaussian noise, quotes kept inside a trust region $|d_t| \le r$, and stable blind retraining, $m < 1$. Two choices are critical to validity. First, cost is measured as profit actually lost relative to never experimenting, the quantity a dealer sees on its books. Second, one cell deliberately violates the smoothness assumption, and its deviation from the theory is reported rather than excused.

D. Exploration Design and Estimation

A dealer pricing many bonds faces a profit hill with many directions, steep in some and flat in others. Experimenting along a flat direction costs little profit yet reveals as much about the market, so the cheapest design moves the quote in proportion to $\Gamma_{\mathrm{PO}}^{-1/2}$, the inverse square root of the curvature in each direction. Spreading experiments evenly instead overpays by exactly

$$F = d\,\mathrm{tr}(\Gamma_{\mathrm{PO}})/\left(\mathrm{tr}\,\Gamma_{\mathrm{PO}}^{1/2}\right)^2 \in [1, d] \tag{3}$$

where d is the number of prices and $F = 1$ only when every direction is equally steep. The agent experiments on a three-point design, quoting slightly above, at, and slightly below the anchor, inside a trust region $r \le 0.8$; it fits the response curve (C_0, C_1, c) by least squares and keeps a running confidence interval for $\hat{\varepsilon}$ from its own trades.

E. The SafeD-PerfGD Agent and Evaluation

As summarized in Fig. 1, SafeD-PerfGD repeats four steps each round. It experiments on the three-point design, updates its fit, computes the correction that would move the quote toward the optimum, and applies that correction only if the confidence interval is narrow enough that even its worst case keeps the retraining loop contracting with a safety margin. Otherwise it freezes the correction and lets blind retraining, which is stable on its own, keep running. Evaluation follows a preregistered list of 36 measured-versus-predicted checks. We compare the agent with finite-difference PerfGD [4], zeroth-order profit optimization, a confidence-bounds explorer [9], and blind retraining over 12 seeds; switch off the design, the anchor, and the gate in turn; transplant the theory unchanged to the pricing economy; and instantiate it on the corporate-bond calibration.

IV. Results and Analysis

A. External Validation of the Simulated Market

Before evaluating the price, we check that the simulated market is the real one. As shown in Fig. 2, our implementation of the calibration reproduces the published anchor h^*, response slope ε^*, and modulus m in 10 of 10 cells, with maximum deviations of 0.23%, 0.62%, and 0.76%, across anchors from 0.435 (investment-grade, calm) to 4.167 (high-yield, crisis). On these cells the exchange rate spans 255.3 to 1.2, and the gap between the anchor and the optimum is 1.9–4.3% of the quoted spread: the money left on the table that the rest of the paper prices.

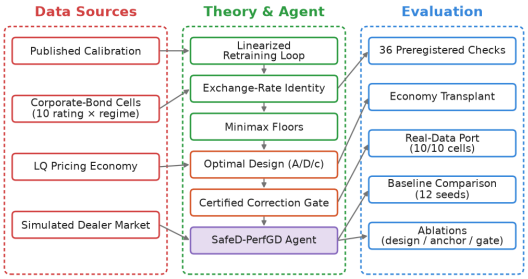


Fig. 1. Pipeline architecture for the proposed pricing framework and certified agent followed by evaluation and benchmarking

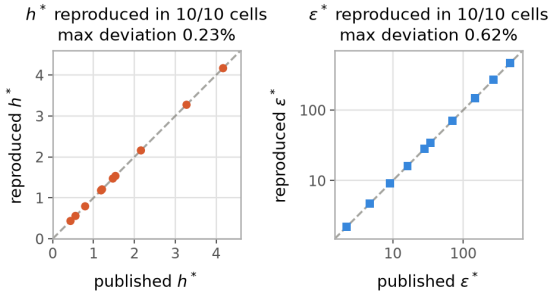


Fig. 2. External validation of the simulated market against a published corporate-bond calibration (10 cells)

B. The Exchange Rate: Global Verification

As shown in Fig. 3 and TABLE I, the measured product of uncertainty and cost sits on the predicted constant in every cell the theory covers: 0.01062 and 0.01070 at exploration amplitudes 0.05 and 0.15 against a predicted 0.01056 at modulus 0.3, and 0.00549 and 0.00569 against 0.00535 at modulus 0.6. Tripling the amplitude moves the product by under 1% at the low modulus and about 3.5% at the high one, well inside the preregistered 15% tolerance. The one cell that violates the smoothness assumption drifts to 0.00622, 16.3% above the constant, so the 36 rows close as 35 passes and one out-of-scope drift cell. These tests together establish the paper's central dissociation: how hard a dealer experiments changes how fast it learns and how much it pays, but not the exchange rate between the two.

C. Where the Price Can Be Lowered: Shape, Not Amount

Because the rate is fixed, the question a dealer controls is where to experiment, not how much. As shown in the left panel

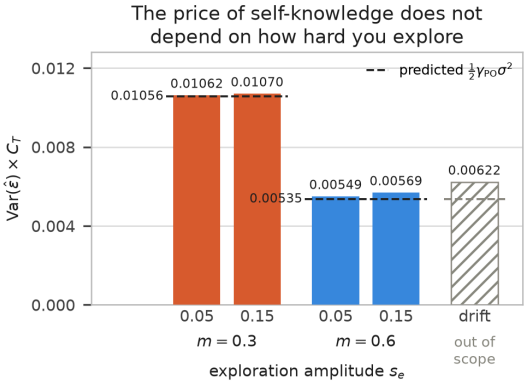


Fig. 3. Exchange-rate invariance across exploration amplitudes and moduli, with the out-of-scope drift cell shown for completeness

TABLE I
Selected Preregistered Verification Rows (36 rows, 0 failures)

Claim	Measured	Predicted
Exchange rate, $m = 0.3$	**0.01062, 0.01070**	**0.01056**
Exchange rate, $m = 0.6$	0.00549, 0.00569	0.00535
Even vs. optimal exploration, $d = 8$	1.529	$F = 1.506$
Best design within reach vs. floor	1.0005	≥ 1
Design beyond reach vs. floor	0.8517	< 1
Pricing economy, $g = 0.4, 0.7$	0.384, 0.192	0.384, 0.192

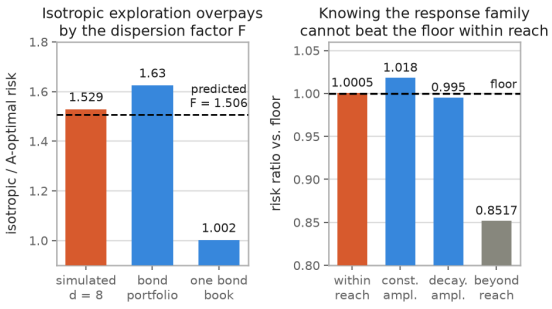


Fig. 4. Where the price can be lowered: the penalty for even exploration, and the reach within which no design beats the floor

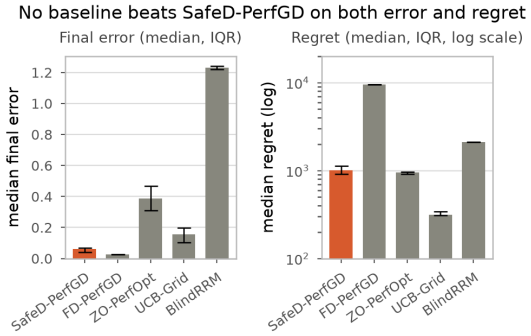


Fig. 5. Baseline comparison over 12 seeds (medians with interquartile ranges)

of Fig. 4, spreading experiments evenly across eight prices costs 1.529 times the optimal shape, against the predicted $F = 1.506$. On the corporate-bond calibration the penalty is 1.63 across the ten-cell portfolio but only 1.002 within one bond: shaping matters when many bonds are priced together, not for a single one. The right panel of Fig. 4 asks whether a dealer who knows the shape of the response curve can beat the rate. Within the reach, two curvature lengths ($2/c$) of the anchor, the best such design comes within 0.05% of the floor (1.0005), and varying the experiment size over time does no better (1.018 and 0.995). Beyond the reach the floor breaks (0.8517), which is why the agent's trust region stays inside it. The rate also settles whether to experiment at all: learning pays off when the gap to the optimum is large, the noise is small, and the dealer is patient.

D. Certified Correction

Knowing the cheapest experiment is not enough; the dealer must also know when its estimate is safe to use. On the live pipeline SafeD-PerfGD settles at $h = 1.258$ against the optimum $h_{\mathrm{PO}} = 1.296$, a 3.0% gap inside the preregistered 8% band, and it never leaves the trust region, with its largest step 0.88 of the cap. The gate opens after roughly 1,300 deployments, a third of the run, during which blind retraining keeps the loop stable while the confidence interval narrows. That waiting time is the regret payment the exchange rate demands. What the dealer deploys is therefore the certificate, a guarantee that the loop keeps settling, rather than the raw estimate.

E. Baseline Comparison

As shown in Fig. 5 and TABLE II, over 12 seeds no baseline beats SafeD-PerfGD on both median final error and median regret. Finite-difference PerfGD [4] lands closer to the optimum, 0.0263 against 0.0599, but pays 9586.79 in regret against 1007.86, more than 5 × as much, because its experiments are never priced. The confidence-bounds explorer [9] pays the least, 315.92, but stops at a final error of 0.1547; it beats SafeD-PerfGD outright in one seed of twelve. Zeroth-order optimization pays slightly less, 952.33, at a final error of 0.3868, and blind retraining trails on both, 1.2293 and 2115.54.

TABLE II
BASELINES OVER 12 SEEDS (MEDIAN)

Method	Final error	Regret
SafeD-PerfGD	**0.0599**	**1007.86**
FD-PerfGD	0.0263	9586.79
ZO-PerfOpt	0.3868	952.33
UCB-Grid	0.1547	315.92
BlindRRM	1.2293	2115.54

F. Ablation Analysis

Ablations. Switching off the design, the anchor, and the gate in every combination, Fig. 6 reveals that the anchor carries the accuracy and the gate carries the safety. With the anchor, final error is 0.0437 under the optimal design and 0.0182 under random jitter, gate or no gate; without it, error rises to between 0.0939 and 0.2825 and the quote never settles. The gate changes discipline rather than accuracy: gated runs freeze for 0.338 to 0.680 of the horizon, ungated runs never do. The design buys knowledge of the curve's shape c: random jitter leaves it 7.8 times more uncertain.

Robustness. The theory survives a change of economy and of modulus. In the simpler pricing economy the exchange rate is exact in all four cells (TABLE I), 0.384 at gain $g = 0.4$ and 0.192 at $g = 0.7$, and the transplanted agent reaches that economy's optimum at 12.65 against 12.5. A loop that never experiments reveals only a bounded amount, because retraining itself jiggles the quote: the predicted cap holds at moduli 0.3, 0.6, and 0.85 to within 6.3%.

Taken together, the results form a single chain: a validated market (Fig. 2) establishes that uncertainty times cost is a constant of the market, unchanged by how hard a dealer experiments (Fig. 3, TABLE I); where to experiment therefore becomes the binding decision (Fig. 4); the certified agent is

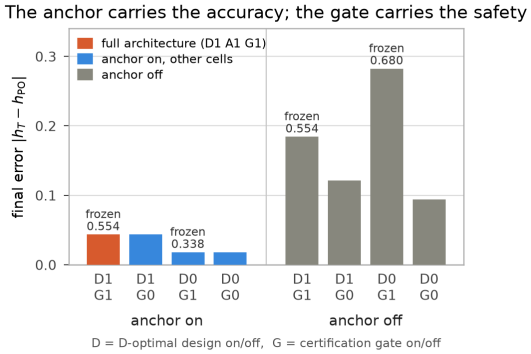


Fig. 6. Ablation of the design, anchor, and gate on final error, with freeze fractions annotated for gated cells

the only one competitive on both error and regret (Fig. 5, TABLE II), with accuracy traceable to the anchor and safety to the gate (Fig. 6); and the rate holds across economies and moduli (TABLE I).

V. Conclusion

Financial dealers must learn how customers respond to their quotes without sacrificing unnecessary profit or destabilizing the market. This paper quantifies the cost of experimentation and identifies how dealers can learn more efficiently while maintaining stability.

We find that the cost of learning is a market constant, $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$, with measured values of 0.01062 and 0.01070 closely matching the predicted 0.01056. Because information gained and profit lost grow at the same rate, dealers can experiment most efficiently along flatter regions of their profit curves. Across 36 preregistered checks, there are zero failures, and the certified agent reaches the optimum without stability violations. No baseline outperforms it on both prediction error and regret.

These findings provide a principled way for dealers to improve their models while minimizing lost profit and ensuring safe deployment.

VI. Limitations and Future Work

While the framework achieves close agreement with its closed forms, certain constraints and assumptions merit discussion.

The results hold near the operating point. Large experiments break the identity, as the one out-of-scope cell shows with its 16.3% drift, and beyond two curvature lengths the floor can be beaten, so the trust region is essential. One of the two lower bounds is proved only to within a factor 13.5, with the exact constant supported by simulation but not yet proved. The safety guarantee is only as good as the confidence intervals it relies on, and a conjectured bound on how long certification takes has a proof strategy only.

Additionally, the corporate-bond leg validates the machinery rather than the size of the effect: the informed-flow channel is set by assumption, and the calibration is reproduced against published values rather than trade-level data.

Future work will tighten the remaining constant, extend the floors to markets with several competing dealers, make the certificate robust to a misspecified response curve, and test the framework on live order-flow data.

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